

RSU VPASS: System Architecture, Manuscript Diagrams & Technical Flow Guide

Project Title: RSU VPASS — A Web-Based Vehicle Registration and Pass Management System with Mobile-Responsive Gate Security Monitoring

Target Institution: Romblon State University (Main Campus, Odiongan, Romblon)

Target Unit: Physical Assets and Security Office (PASO)

Document Purpose: Manuscript Documentation, Capstone Defense Preparation, and AI / Agent Engineering Handoff

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1. Executive Summary & Core Institutional Rationales

1.1 Overview

RSU VPASS is an integrated full-stack web platform built for Romblon State University to replace manual paper vehicle registration, eliminate physical office-to-office clearance queues, and equip campus gate guards with mobile-responsive scanning and visitor monitoring tools.

1.2 Four Key Architectural & Operational Decisions (Crucial for Defense & Manuscript)

IMPORTANT

1. Downloadable & Client-Installed Gate Pass Stickers (Institutional Relief)

- **The Problem It Solves:** In traditional setups, PASO must budget, procure materials, and print thousands of physical plastic/decal stickers every single school year. This causes recurrent financial expenses, logistical delays, and supply shortages.
- **The VPASS Solution:** Upon cashier payment approval, the system generates an official high-resolution gate sticker with an encrypted QR code directly in the client's portal. Clients print and install the sticker onto their vehicle windshield or motorcycle chassis.
- **Anti-Tampering Security:** Physical paper does not hold the security; the security is rooted in the database-backed encrypted QR code (`RSU-VPASS:<PassNumber>:<PlateNumber>:<SchoolID>`). Photocopying a sticker or altering the printed text fails immediately upon guard scan because the scanner pulls authentic registration data directly from the server.

NOTE

2. Student Traffic Strategy: Visual Inspection & Spot-Checking (No Gate Bottlenecks)

- **Traffic Reality:** During peak hours (7:30 AM – 8:30 AM), hundreds of student motorcycles enter through campus gates simultaneously. Scanning every single student motorcycle would back up queues onto the national highway, creating road safety hazards.
- **Operational Flow:** Students with valid displayed stickers pass through smoothly without stopping. Security guards only stop and scan student vehicles during **periodic spot-checks, random inspections, or when observing suspicious or unstickered vehicles.**

TIP

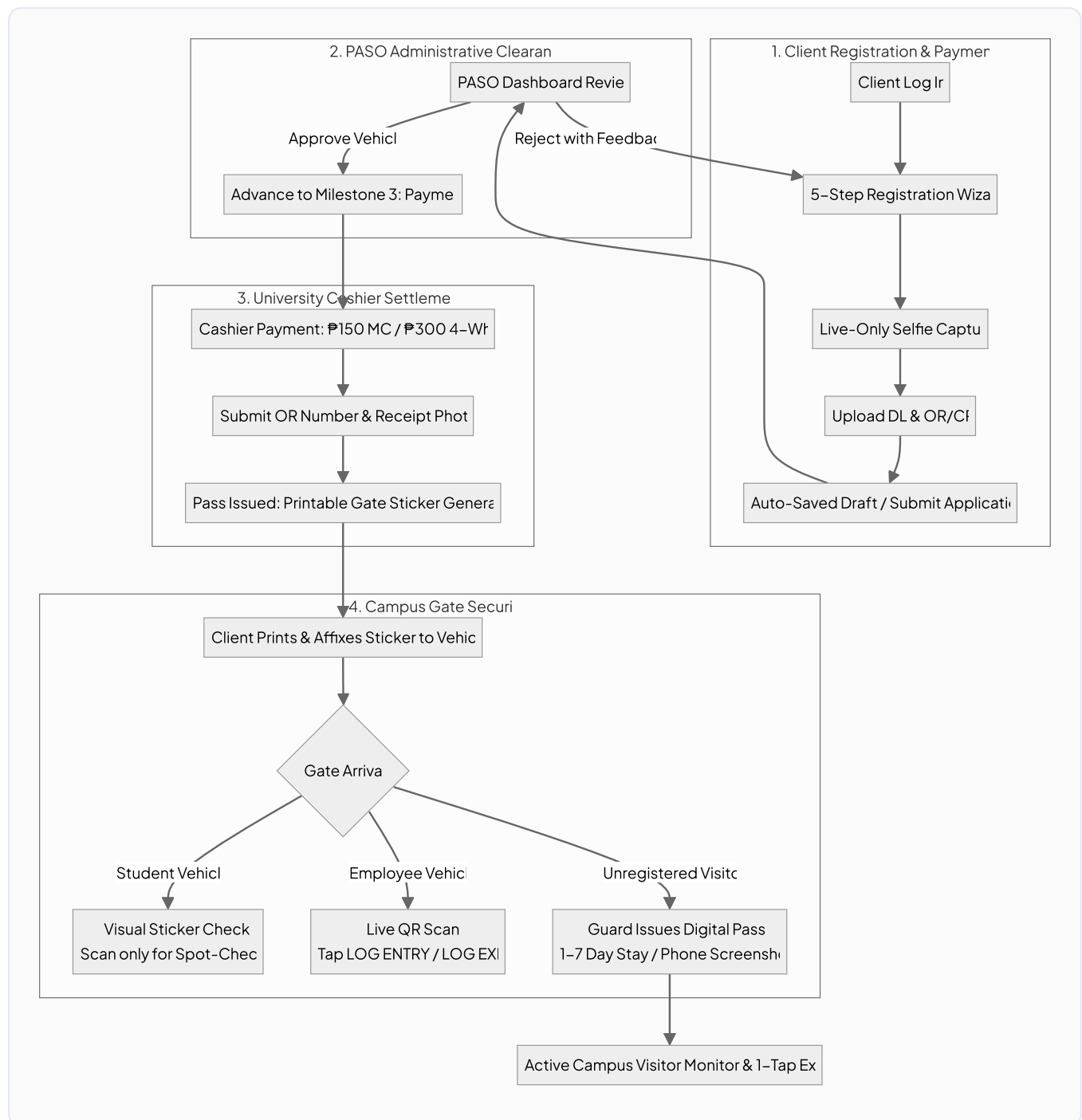
3. Dedicated Employee Entry and Exit Logging (Institutional Accountability)

- For university faculty and staff, PASO and HR administration require verifiable records of personnel presence during official working hours.
- When an employee pass is scanned, the guard scanner interface displays dedicated touch buttons: `[ LOG ENTRY ]` and `[ LOG EXIT ]`, recording timestamps, gate location, plate number, and the guard on duty.

IMPORTANT**4. 100% Digital Temporary Visitor Pass (Single-Day & Multi-Day Event Stays)**

- **Zero Physical Stickers:** Visitors are on campus temporarily; printing physical stickers for short-term visitors wastes university resources.
- **Configurable Duration:**
 - **1 Day (Default):** Automatically expires at 11:59 PM on the day of issuance (for couriers, parents, short errands).
 - **Multi-Day (2, 3, 5, 7 Days / 1 Week):** Designed specifically for university delegates, sports meets, accreditation teams, and seminar guests so they do not need to register every day.
- **Paperless Screenshot Pass:** The guard issues the pass in 15 seconds; a modal displays the pass with an encrypted QR code and plate number for the visitor to photograph/screenshot on their smartphone.
- **Active Tracking & 1-Click Renewal:** Guards monitor an "Inside Campus" tab and tap **LOG EXIT** upon departure. For returning visitors, a **Renew Pass** button pre-fills their details with 1 click.

2. End-to-End System Workflow Narratives



Detailed Workflow Steps:

1. Client Registration:

- The user logs in via Student ID / Employee ID.
- Opens "+ Register New Vehicle" launching the 5-Step Milestone Wizard.
- Enforces **Live-Only Webcam Capture** (front camera with oval alignment guide; gallery uploads disabled to prevent spoofing).

- Any progress is auto-saved to browser `localStorage` (`vpass_draft_application`). If closed, a "Resume Draft" banner appears.

2. PASO Administrative Clearance:

- PASO officers view pending applications, examine Driver's License and OR/CR documents, and either approve or reject with written feedback.

3. Cashier Payment:

- Approved requests move to Milestone 3 (Cashier Payment).
- Motorist pays official parking fee at Cashier (Motorcycle: ₱150, 4-Wheels: ₱300).
- Motorist inputs OR Number and uploads receipt photo.

4. Pass Generation & Self-Installation:

- Milestone 4 unlocks: "Pass Issued".
- Client downloads high-resolution gate sticker with QR code, prints it, and adheres it to their windshield or motorcycle chassis.

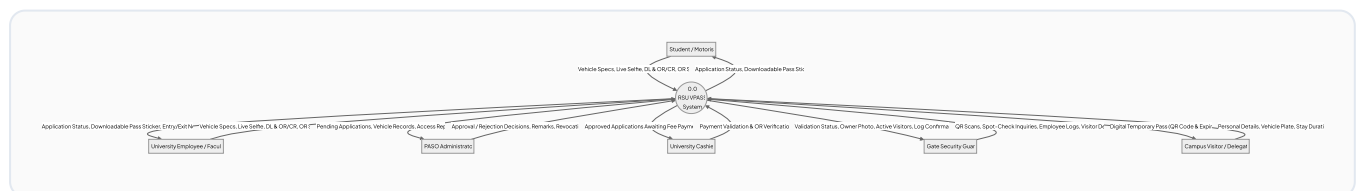
5. Gate Security & Visitor Management:

- Students:** Quick visual check; scanned only for spot-checks / suspicious inspection.
- Employees:** QR scanned; guard selects `LOG ENTRY` or `LOG EXIT`.
- Visitors / Event Delegates:** Guard enters name, plate, destination, selects duration (1 to 7 days). Instant digital pass is displayed for the visitor to photograph/screenshot. Active visitor list tracks departure via 1-tap exit.

3. Manuscript Diagrams (With Mermaid & Text Content)

3.1 Context Diagram (Level 0 DFD)

The Context Diagram defines the system boundary, showing all external entities and their bidirectional information flows with the **RSU VPASS System**.

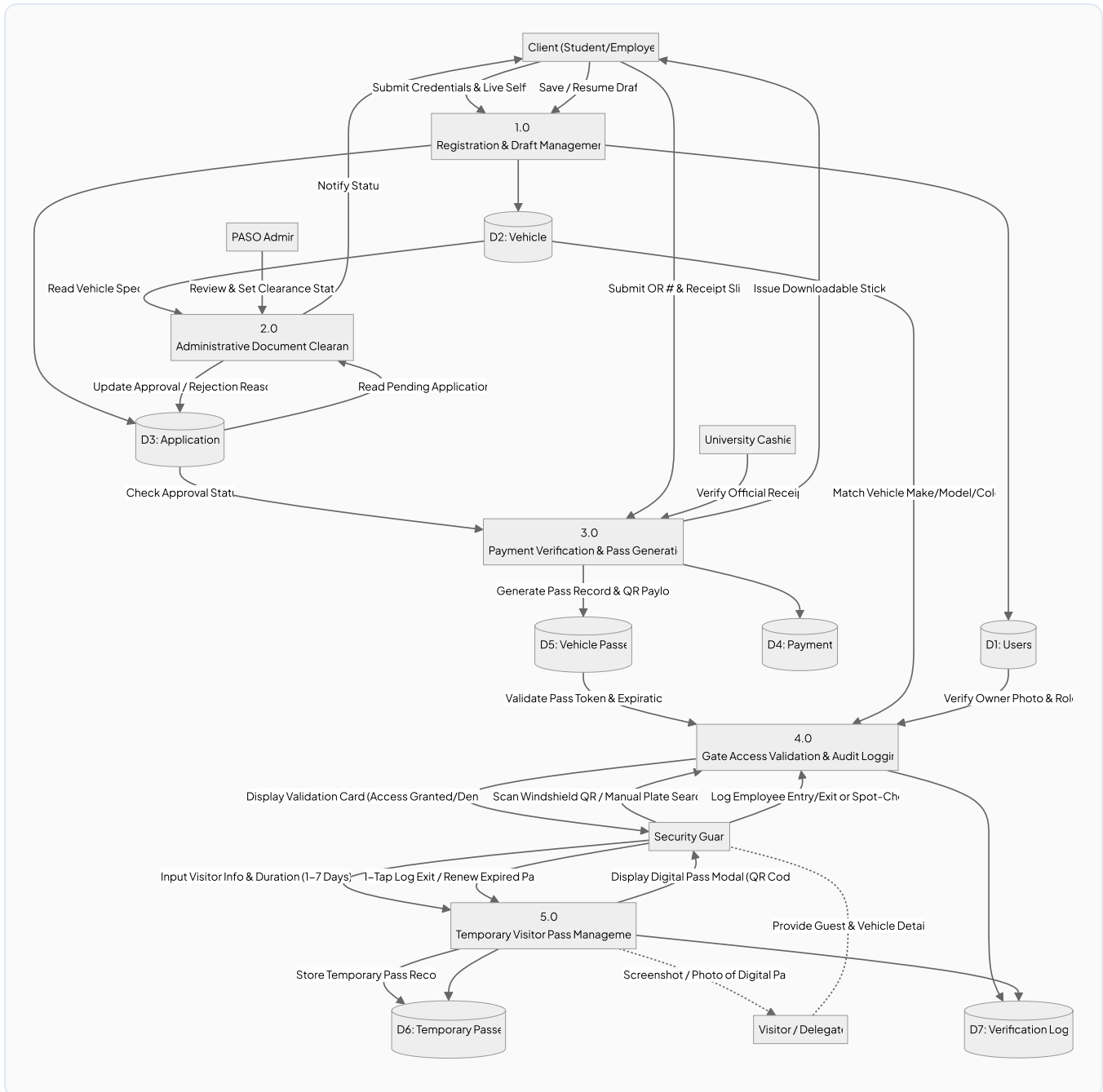


Content Table for Manuscript Description:

External Entity	Data Flow to System (Inputs)	Data Flow from System (Outputs)
Student / Motorist	Vehicle details, live selfie, Driver's License & OR/CR scans, Cashier OR # and slip photo.	Registration status updates, draft restore, downloadable vehicle gate pass sticker.
University Employee	Vehicle details, live selfie, official credentials, Cashier OR # and slip photo.	Registration status, downloadable vehicle sticker, real-time entry/exit logging feedback.
PASO Administrator	Clearance decisions, rejection reasons, clearance revoking, audit report requests.	Applicant queue, verified vehicle documents, campus security analytics, gate logs.
University Cashier	Fee payment verification, Official Receipt (OR) issuance.	Fee billing schedule (Motorcycle ₱150, 4-Wheels ₱300), payment queue.
Gate Security Guard	QR scan data, spot-check plate searches, Employee Entry/Exit logs, Visitor pass inputs.	Verification card (Access Granted/Denied, owner photo), active visitor list, audio/haptic alert.
Campus Visitor / Delegate	Guest name, plate number, contact number, host destination, requested duration (1–7 days).	Digital Temporary Pass with QR code, expiration timestamp, entry authorization.

3.2 Data Flow Diagram (DFD Level 1)

Deconstructs the main system into 5 discrete sub-processes with 7 primary data stores.



3.3 Use Case Diagram

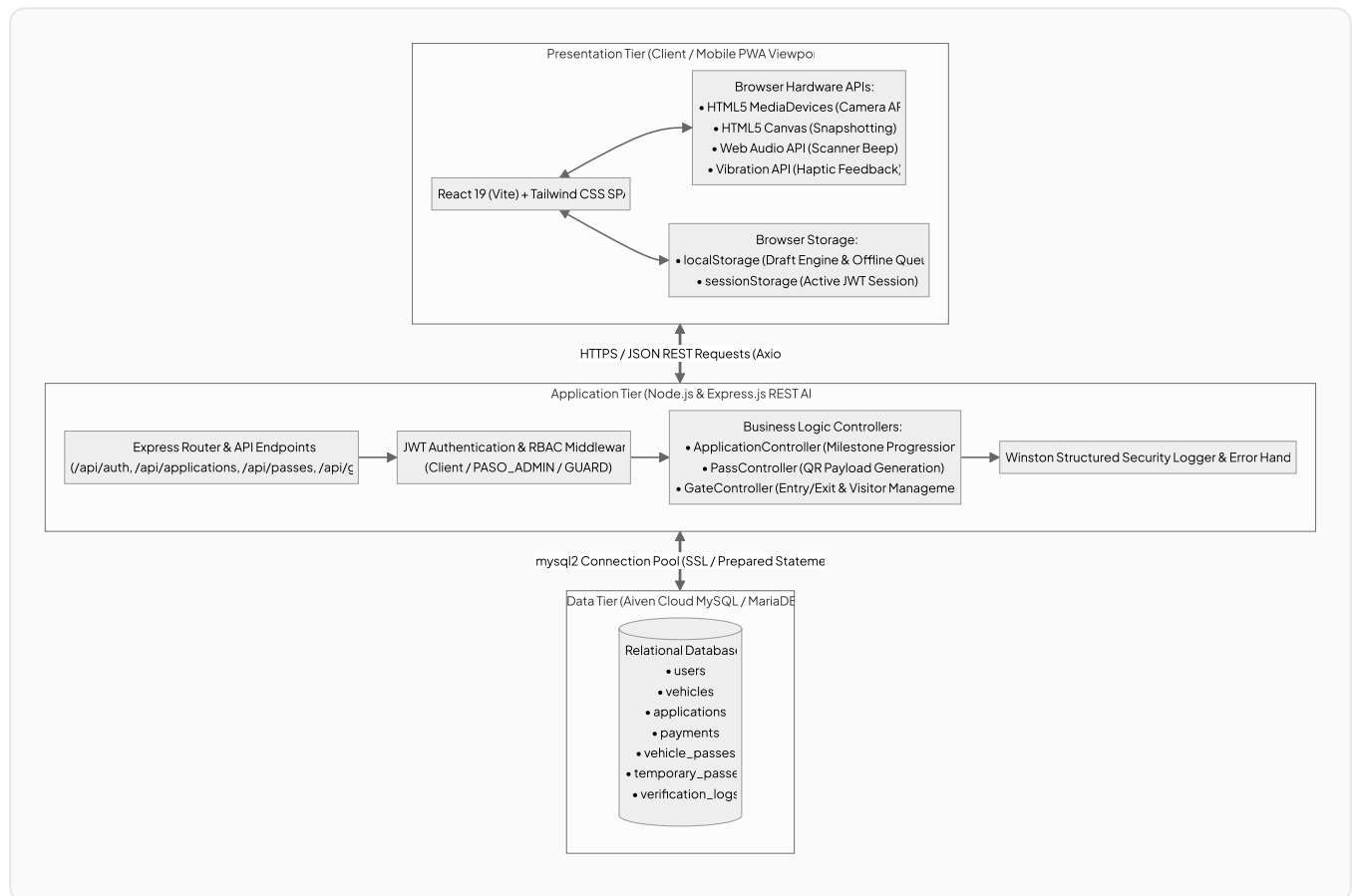
Highlights the role-based functional capabilities across the four primary actors: **Client (Student/Employee)**, **PASO Administrator**, **Gate Security Guard**, and **System/Cashier**.



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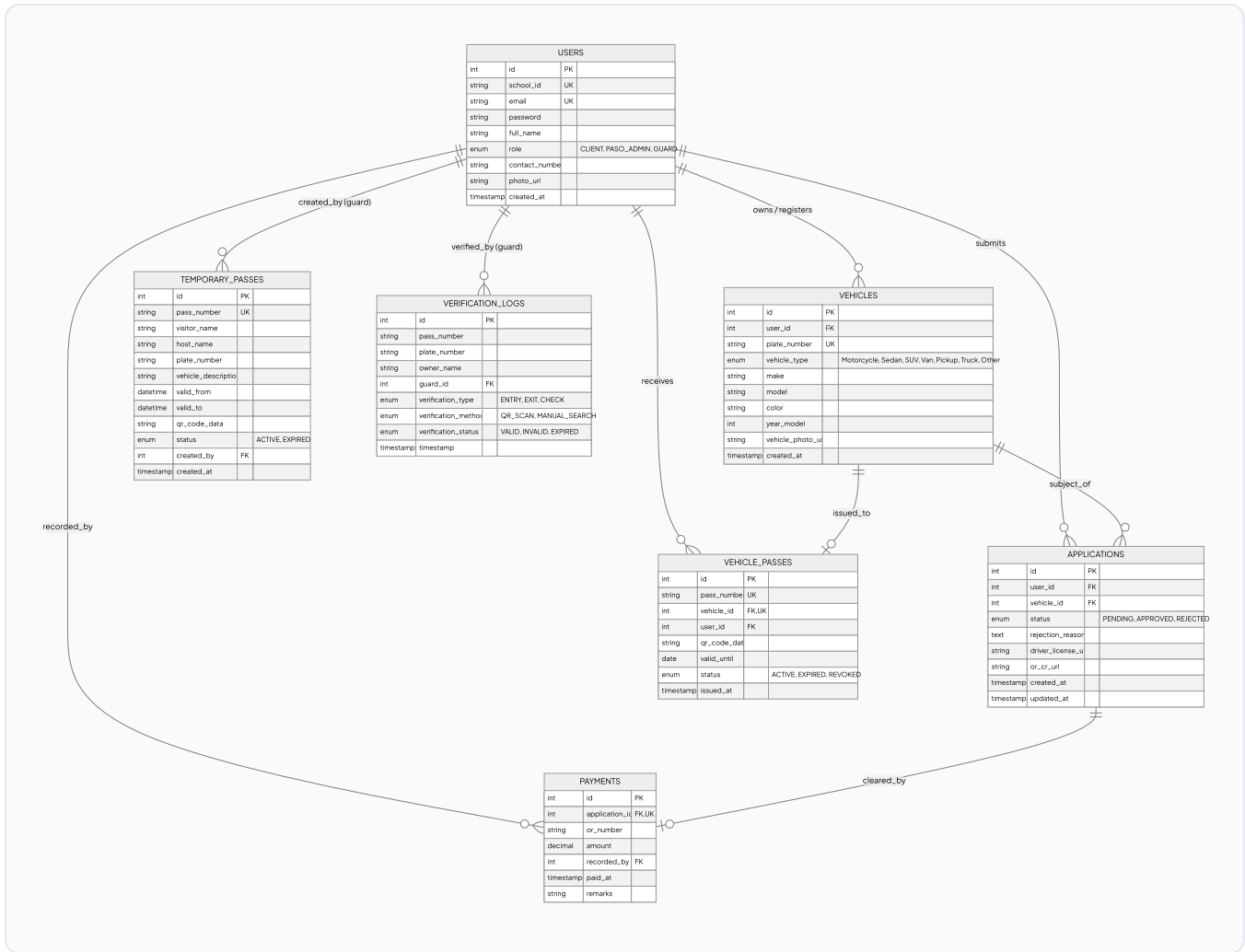
3.4 System Architecture Diagram (3-Tier)

Illustrates the physical architecture, hardware API interactions, application server layer, and cloud database layer.



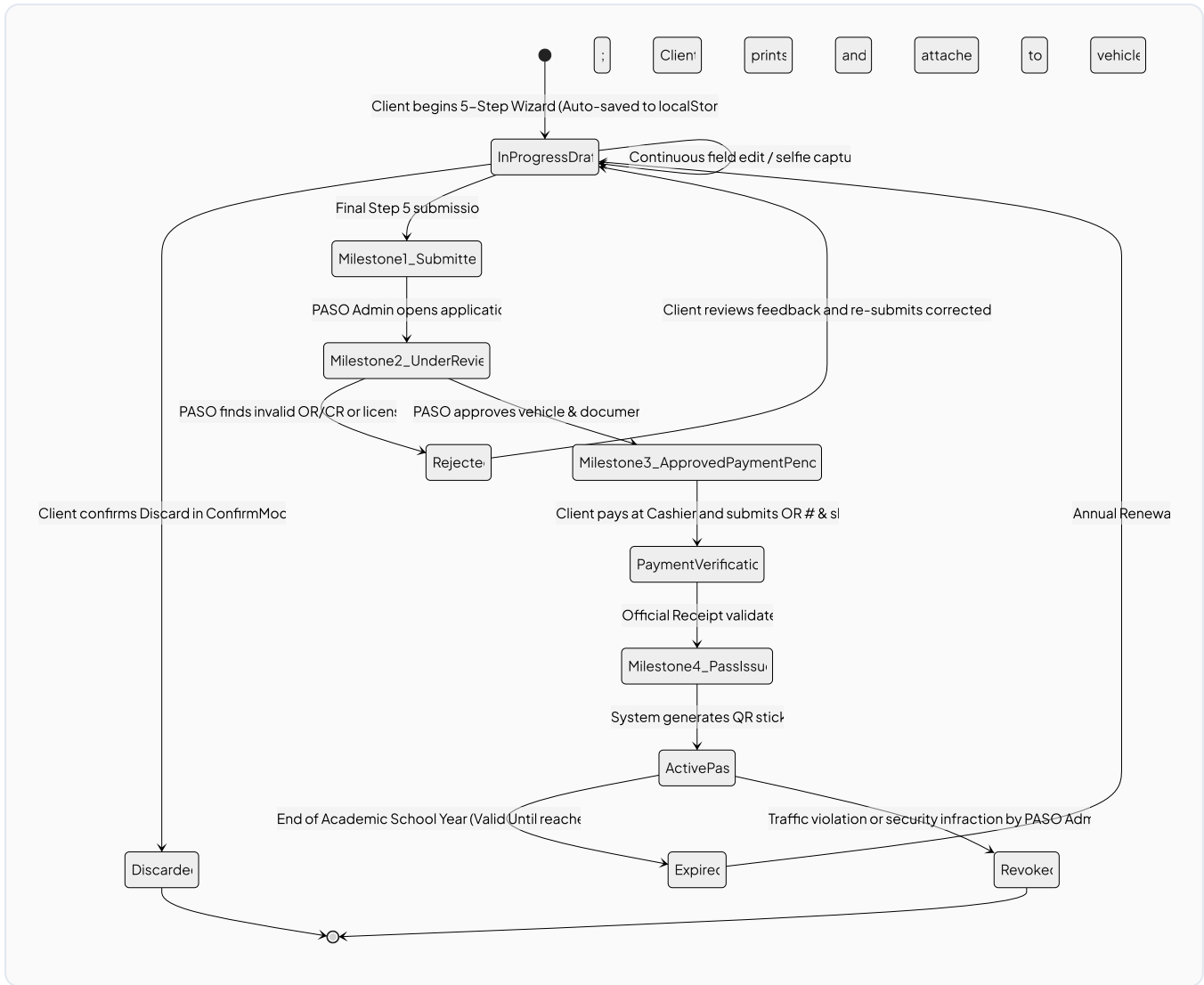
3.5 Entity Relationship Diagram (ERD)

Matches the exact database schema implemented in `server/database/schema.sql`.



3.6 State Transition Diagram (Application & Pass Lifecycle)

Illustrates the exact lifecycle states of an application from initial drafting to active pass status and expiration.



4. AI & Developer Handoff Reference (Future Revisions Guide)

This section provides future AI agents and software engineers with exact code locations, state conventions, and implementation rules.

4.1 Key File Sitemap

File Path	Role & Content
<code>client/src/App.jsx</code>	Main application routes, role-based route guards (<code>ProtectedRoute</code>), and context wrapping.
<code>client/src/pages/guard/GuardScanner.jsx</code>	Mobile gate scanner: HTML5 camera QR reader, spot-check manual search with classification filters, dedicated Employee Entry/Exit buttons, 1-to-7 day Digital Temporary Visitor Pass modal, Active Visitor Monitor, and 1-click renewal.
<code>client/src/pages/client/MyVehicle.jsx</code>	5-step vehicle registration wizard, live webcam selfie capture (no gallery upload), auto-saving draft engine, and <code>ConfirmModal</code> for discard safety.
<code>client/src/pages/client/VehiclePass.jsx</code>	Official downloadable & printable gate pass sticker generator with encrypted QR code.
<code>client/src/pages/client/Applications.jsx</code>	4-stage milestone tracker (Submitted → PASO Review → Cashier Payment → Pass Issued).
<code>client/src/pages/admin/AdminDashboard.jsx</code>	Desktop PASO admin portal: Document inspection viewer, Approve / Reject with remarks.
<code>server/database/schema.sql</code>	MariaDB / MySQL DDL schema and demo seed accounts.

4.2 Local Storage & State Contracts

- `vpass_draft_application` : Stores in-progress registration drafts (steps 1–5, field values, live photo base64, uploaded files).
- `vpass_visitor_passes` : Stores temporary visitor pass records (pass number, visitor name, vehicle plate, duration, issued at, valid until, status: `INSIDE_CAMPUS` vs `DEPARTED`).
- `vpass_token` : User authentication JWT bearer token.
- `vpass_user` : Active user profile object with role enum (`CLIENT` , `PASO_ADMIN` , `GUARD`).

4.3 QR Code Payload Encryption Protocol

Vehicle passes generate a standard encrypted token formatted as:

```
RSU-VPASS:<PassNumber>:<PlateNumber>:<SchoolID>
```

Example: `RSU-VPASS:VP-2026-0001:XYZ5678:2026-00001`

Visitor Pass Example: `RSU-VPASS-TEMP:VP-TEMP-20260923-ABCD:XYZ1234:VISITOR`

4.4 Rules for Future Revisions

1. **Never re-enable gallery file uploads for selfies:** The live front-facing webcam capture is an anti-spoofing requirement.
2. **Never force 100% scanning on all student entries:** Student entry must remain smooth visual inspection with spot-checking to avoid highway traffic congestion.
3. **Never require physical stickers for visitors:** Visitor passes must remain 100% digital with screenshot/photo capability.
4. **Visitor Duration Flexibility:** Any new UI changes to the visitor form must preserve the multi-day selector (1, 2, 3, 5, 7 days) to support school delegates and conferences.